

The Spy Within



Unknown to most PC users, information is stolen from within our PCs every day. Personal details, knowledge of our surfing habits, even passwords and credit card information are at high risk of being stolen. How, you ask? Spyware—small applications in our own computers which collect and push the information to the creator of the application.

What is Spyware?

Renowned security expert Steve Gibson of GRC.com describes spyware as “any software which employs a user’s Internet connection in the background—the so-called backchannel—without their knowledge, or explicit permission. Silent background use of an Internet backchannel connection must be preceded by a complete and truthful disclosure of proposed backchannel usage, followed by the receipt of explicit and informed consent for such use. Any software communicating across the Internet, and missing

these elements, is guilty of information theft and is properly and rightfully termed: Spyware.” This broad definition includes Trojans, remote administration tools, key loggers and other similar hacker tools. How bad, or dangerous a particular piece of spyware is, depends on three things—what type of information it gathers, what the company collecting the information does with it and if the user’s

consent was taken before collecting this information.

The information gathered could be benign, such as your surfing habits and interests, or it could be far more important, such as personal details and sensitive data. Information about your interests and the sites you’ve visited can be used for your benefit, so that advertisers show advertisements that you find interesting or recommend purchases that would appeal to you. These tools which display targeted advertising, based on the information they collect about you, are called Adware, and are the most common form of spyware. The Gator Advertising and Information Network (GAIN) is a very good example of adware. The company proudly claims to have 35 million users who advertisers can reach directly in the form of pop-up ads. Gator.com though, are open enough to admit they also deliver advertisements, information and software, based on the Web sites you view.

Some browser cookies, also known as Tracking Cookies, can be considered as the most basic form of spyware—especially when Web sites store your personal details, without your knowledge, and the same is accessible to others. Normal cookies are mostly harmless, and almost always improve a user’s Internet experience. Simple examples are cookies that remember login information, or message boards that show you what’s new since the last time you visited. A major distinction here is that tracking cookies are not hidden and the user has the option to disable them along with other cookies.

Key loggers and Trojans usually have a more sinister use. Key loggers are small applications that run in the background and record every key that you type. They can collect a wealth of information: passwords, credit card details, private e-mails, chat sessions... everything. This information is stored locally in a log file, and can be silently sent out via e-mail. Trojan’s are small programs that let a remote user take complete control of your PC, and perform some impressive feats, such as monitoring your system, scanning your drives, stealing files and even installing tools such as key loggers. Fortunately Trojans are removed by most anti-virus tools today, and have since, become less common.

But I never installed any spyware!

Most of the spyware you find on your home PC has been installed by you, albeit unknowingly, along with some other application. Popular peer-to-peer file sharing utilities, such as KaZaA and Morpheus, which are distributed free to users, install adware elements when you install the program. Companies do this to support the large development costs associated with making a good application and keeping it free. In effect, the adware pays for the development, and you get the applications without paying, but not without a price—the loss of your privacy.

These companies are quite blatant about their activities. When questioned, they simply ask users to “read the fine print”; the End User Licence Agreement (EULA)—the bit that we don’t read, and ‘Agree’ to without hesitation. These EULAs are very shrewdly written, so as to conceal from the user the fact that his, or her personal information will be collected.

Even more sinister are drive-by downloads. These are tools which automatically install themselves when you visit a Web site or click on a misleading link. They usually install themselves as toolbars, add a few shortcuts to your browser’s Favorites list, and make themselves your home page. Even if you change the



GAIN’s ads indicate that they are from Gator